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Review article

Multispectral imaging: Review of current applications



Feiyan Ma, MD, PhD ^{a,*}, Mingzhen Yuan, MD ^{b,1}, Igor Kozak, MD, PhD ^{c,**}

^a The Second Hospital of Hebei Medical University, Ophthalmology Department, Shijiazhuang, China

^b Beijing Tongren Hospital of Capital Medical University, Ophthalmology Department, Beijing, China

^c Moorfields Eye Hospitals UAE, Abu Dhabi, United Arab Emirates

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ABSTRACT

Multispectral imaging (MSI) is a unique layer-by-layer imaging technique that allows the visualization of a wide array of retinal and choroidal pathologies including retinovascular disorders, retinal pigment epithelial changes, and choroidal lesions. Herein, we summarize the basic imaging principles and current applications of MSI together with recent technology advances in the field. MSI detects reflectance signal from both normal chorioretinal tissue and pathological lesions. Either hyperreflectance or hyporelectance reveals the absorption activity of pigments such as hemoglobin and melanin and the reflection from interfaces such as the posterior hyaloid. Advances in MSI technique include creation of a retinal and choroidal oxy-deoxy map that could provide a better understanding of blood oxygen saturation within lesions as well as better interpretation of reflectance phenomenon of MSI images such as the different reflectance from the Sattler and Haller layers described in this review.

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1. Introduction

Multispectral imaging (MSI), generally speaking, is the application of full-spectrum imaging that could yield improved ability to distinguish different chromophores in the tissue.³⁶ It can capture images at various wavebands and associate spectral signatures with chemical components by absorbing light frequencies that resonate with the chemical bonds. MSI has now been applied in precision agriculture, artwork

restoration, environmental monitoring, and biochemistry.^{3,11,12,39} MSI in ophthalmology is based on a light emitting diode (LED) imaging system that integrates discontinuous narrow band spectrum light sources and results in a series of grayscale images based on the absorption and reflection of anatomical and pathological structures of the posterior segment.²⁸

Traditionally, fundus fluorescein angiography (FFA) and indocyanine green angiography (ICGA) have been used as diagnostic imaging studies for various retinal and choroidal

* Corresponding author: Feiyan Ma, MD, PhD, The Second Hospital of Hebei Medical University, 215 Heping Rd, Shijiazhuang, China.

** Corresponding author: Igor Kozak, MD, PhD, Moorfields Eye Hospitals UAE, Abu Dhabi 62807, United Arab Emirates.

E-mail addresses: mafeiyang2004@163.com (F. Ma), igor.kozak@moorfields.ae (I. Kozak).

¹ The authors contributed equally.

disorders; however, allergic reactions have been reported for the intravenous fluorescence dyes ranging from nausea, dry cough, rhinorrhea to severe anaphylactic shock including drop in blood pressure as well as loss of consciousness.⁸ MSI is a noninvasive dye-free imaging test for chorioretinal disorders with high diagnostic power. For example, sensitivity and specificity of MSI in the diagnosis of diabetic retinopathy (DR) is 97.4 and 100% compared with FFA.²⁶ Sensitivity and specificity of MSI in diagnosis of polypoidal choroidal vasculopathy (PCV) are 84.21% and 93.10%, respectively, when using ICGA as diagnostic criteria.²⁹ Such an approach could potentially reduce the necessity of FFA/ICGA use when making a clinical diagnosis.

Autofluorescence (AF) has been considered a useful non-invasive tool in the diagnosis of retinal pigment epithelium (RPE)-related disorders based on the detection of fluorophores in RPE with excitation filters of 476 and 550 nm and two barrier filters at 499–576 nm.³³ MSI is based on the imaging of melanin in RPE at 680 and 780 nm.⁴² FAF yields one slice of gray scale image that displays pathological activities of the RPE. Excess lipofuscin level could lead to increased FAF signal (hyper-AF) and disrupted lipofuscin could present as decreased FAF signal (hypo-AF); meanwhile, MSI could present a set of gray scale images based on pigment matter (including melanin) on different layers of the fundus. FAF has been applied in imaging lipofuscin-related disorders such as Stargardt disease and age-related macular degeneration (AMD)³³; however, FAF may fail to detect nonlipofuscin abnormalities²⁸ in situations where the RPE is not yet disrupted such as retinal vascular disorders or choroidal tumors.²⁸

MSI emerges as a noninvasive imaging technique with advantages of improved visualization of retinal layers that allows for the capture of images at different depths in the retina and provides better visualization of different retinal layers.²⁵ This can aid in the detection and diagnosis of certain retinal conditions such as macular degeneration and DR. MSI has also permitted detection of subtle changes in the retina that may not be visible with traditional imaging techniques,¹⁰ which can be especially useful in monitoring the progression of retinal diseases over time. Reduced need for contrast agents permits MSI images capture without the need for contrast agents, reducing the risk of adverse reactions to contrast agents. The imaging process is a quick and easy experience that can be performed in the doctor's office, potentially saving clinical time and reducing the need for patients to travel to other facilities for imaging tests.

We provide a clinically based summary of the application of MSI in retinal and choroidal diseases.

2. Principles and equipment

2.1. Imaging principles

2.1.1. Principle of tissue penetration depth

Tissue penetration depth refers to the depth at which light can penetrate into biological tissues before being absorbed or scattered.³¹ It is a physical property that is determined by the wavelength of light and the optical properties of the tissue, such as its absorption, scattering, and refractive index. Generally, longer wavelengths of light have greater tissue penetration depth than

shorter wavelengths. Shorter wavelengths (such as light less than 500 nm) are usually scattered by the human lens, and imaging of retina tissue can be disrupted. MSI of the fundus usually involves 3 or more wavelengths that cover visible light over 500 nm and near-infrared (NIR) light. Commercially available MSI can generate a series of gray scale images using a range of monochromatic lights with wavelength from 550 to 850 nm, which are able to visualize the retina from the inner limiting membrane down to the choroid.^{33,42} Imaging at various tissue penetration depths in MSI is demonstrated in Fig. 1A. Detailed relevance of spectrum to sliced layers of the fundus is illustrated in Fig. 2. 550 nm is advantageous in the imaging of vitreoretinal interface. 600 nm penetrates deeper into the neurosensory retina and 680 nm is more focused on the RPE. 780 nm penetrates the choriocapillaris and Haller's layer of the choroid, while 810 nm is more potent in presenting the structure of the Sattler layer of the choroid. MSI and the layers of the fundus are highly related, but they do not directly correspond at all layers of the retina.

2.1.2. Differential light absorption

Differential light absorption is a metric that characterizes the wavelength-dependent light absorption properties of imaged biological tissue.^{24,39} This measure is influenced by the concentration of absorbers that are specific to certain wavelengths within the tissue. MSI reflectance is also related to differential light absorption in the human fundus. There are three main pigments in the fundus: macular pigment, retinal/choroid hemoglobin, and RPE and choroid melanin.²⁷ Chemicals related to different physiology and pathology can be identified by their specific combination of spectral peaks that correspond to the absorbance bands resulting from the chemical bonds resonance with the incident light.²⁷

- (1) *Macular pigment* is composed of two main carotenoids, lutein and zeaxanthin. The peak absorption wavelength for the macular pigment is around 460–470 nm, which was optimal wavelength for the observation for macular pigment.¹³
- (2) *Hemoglobin*: Most MSI cameras utilized wavelengths ranging between 500 and 700 nm to examine the retinal vascular layers and distinguish between oxygenated and deoxygenated blood, similar to conventional red-free images obtained by commercial fundus cameras that use a green filter.³² Absorption is different for oxygenated and deoxygenated hemoglobin because oxygenated hemoglobin absorbs more light at shorter wavelengths (around 450–500 nm), while deoxygenated hemoglobin absorbs more light at longer wavelengths (around 650–700 nm). By analyzing the differences in the light absorption of these two forms of hemoglobin, MSI can provide information about the oxygenation levels of blood vessels in the imaged retina tissue (retinal oxy-deoxy maps).² Dual-wavelength techniques (using LEDs with peaks within NIR) were utilized to enable deeper penetration into the retina due to the lower absorption of melanin and to provide further insights into the oxygenation status of the choroidal vasculature (choroid oxy-deoxy map). Oxy-deoxy maps have been used to detail oxyhemoglobin in the retinal and choroidal vasculatures. Areas of oxygenated blood are hyperreflective (white in

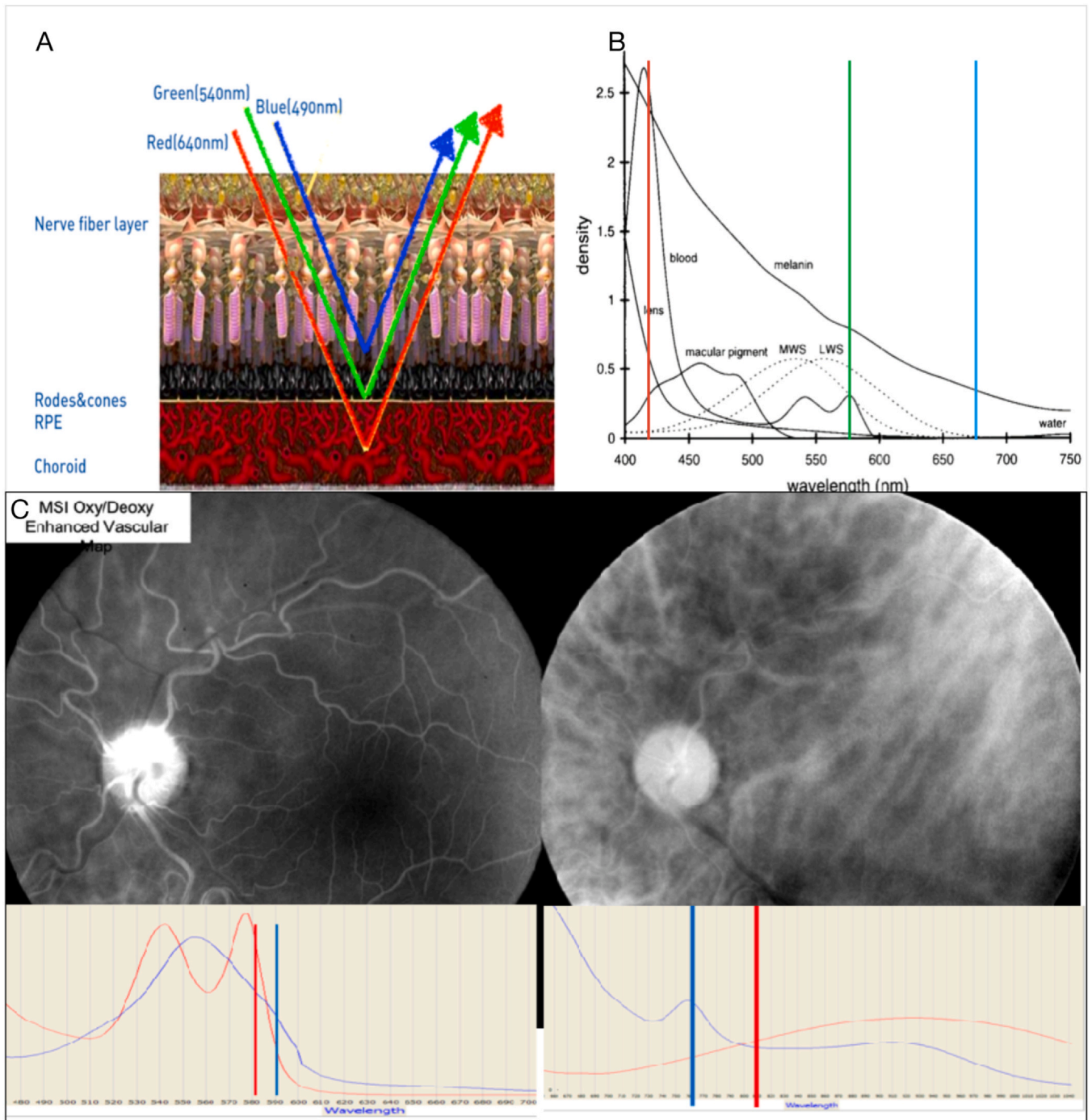


Figure 1 – Principle of multispectral imaging. The longer the wavelengths, the better of the penetration of different spectrum (A). Hemoglobin showed a peak absorption between 550 and 600 nm (green line), and melanin showed more absorption between 650 and 700 nm, which provide a window for observation of the pigment for PRE layer (B). Both middle and long (infrared) wavelengths showed peak absorption around 550 nm. Water showed little absorption over 500 nm. Oxy-deoxy map of the retina and choroid (C left and right respectively).

appearance), while areas of deoxygenated blood are hyporeflective (dark in appearance) implying either non-perfused or hypo-perfused retina.⁴³

- (3) **Melanin:** The absorption of melanin is optimal in the range of 450–600 nm,⁴² but it becomes overshadowed by other prominent ocular components such as hemoglobin, lutein, and rhodopsin in this part of the spectrum. Beyond

600 nm, melanin is the only prominent retinal pigment, and images acquired at these wavelengths can effectively enhance the visualization of retinal structures containing melanin, particularly the underlying RPE. 600–700 nm was chosen for the observation of melanin in RPE in most MSI cameras, while wavelengths over 700 nm were chosen for the revelation of melanin in the choroid.²⁹

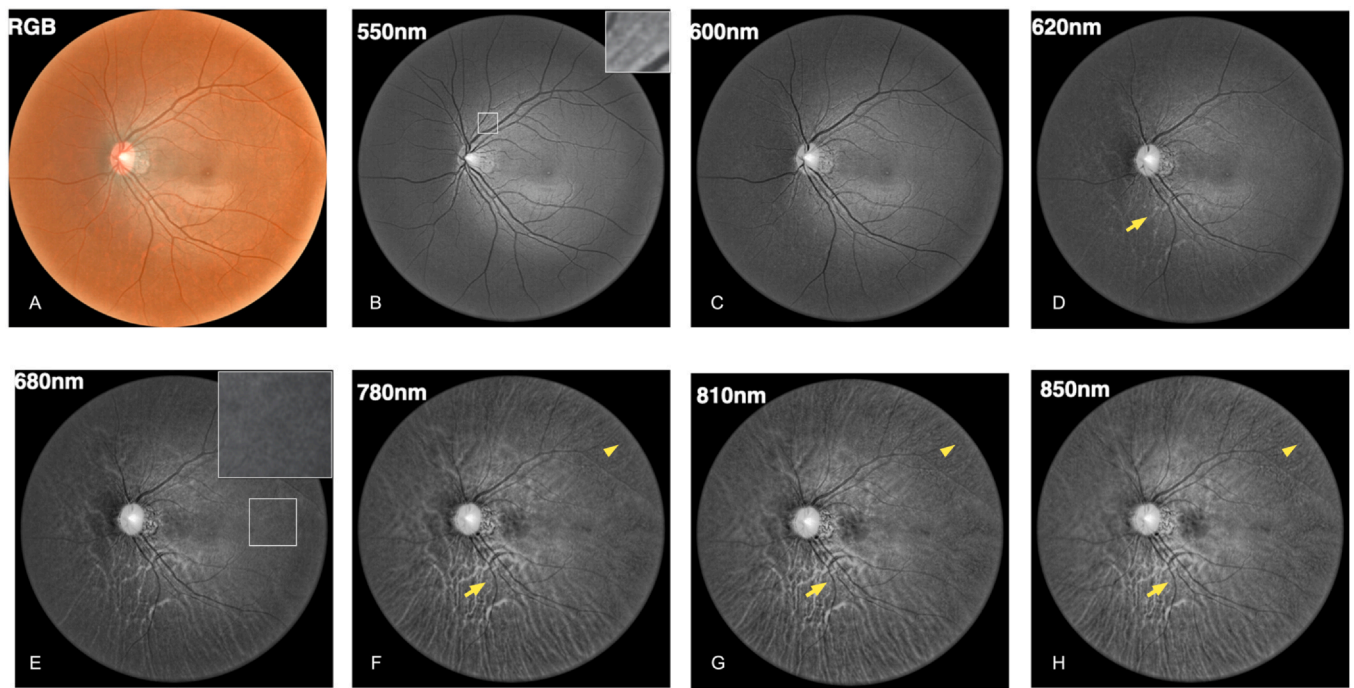


Figure 2 – Multispectral imaging of normal human eye showing retina imaged using sequential wavelengths. A: RGB image; B: 0.550 nm shows inner retinal layers including posterior hyaloid interface around the macula, retinal nerve fiber layer (inset); 600 nm (C) and 620 nm (D) show intraretinal layers with transmission of choroidal vessels in 620 nm (yellow arrow). 680 nm (E) shows outer retinal layers, retinal pigment epithelium and more choroidal vessel transmission. 780 nm (F), 810 nm (G), and 850 nm (H) show the choroid vasculatures. Hyporeflectant linear structures with medium-sized choroidal vessels indicate Sattler's layer (yellow arrowheads on F–H), while large hyper-reflectant linear choroidal vessels indicate Haller's layer, defined as the outer choroidal layer containing the large choroidal vessels.

2.2. Equipment evolution

The first equipment for MSI was based on emitting light with spectrum settings of 507, 525, 553, 584, 596, and 617 nm which were shown theoretically to separate the contributions of each chromophore (Table 2). This system was shown to capture chromophores present in the fundus of the eye in a relatively layered fashion.³⁸ The first commercial MSI system was developed in 2012 (RHA, Annidis, Canada) using narrow-band LED light sources ranging from 520 to 940 nm to create a monochromatic series of en-face images with field of view of 45 degrees.¹⁵ In 2016, the first MSI system with one-click field of view of 63 degrees was introduced (C2000, Shenzhen Thondar, China) with spectra of 550, 600, 620, 680, 780, 810, and 850 nm.²⁸ In a more recent study by Alterini,¹ an MSI camera with wider wavelength ranges from 416 to 1213 nm has been developed to that allows for better investigation for retina, choroid and even, potentially, scleral pathologies.

3. Current application

3.1. Terminology in MSI

Several terminologies have been used in MSI, including clinical terminologies that provide both qualitative and quantitative indicators of chorioretinal structure. Qualitative

indicators include *en face* images of the retina/choroid and its layers. These images can be used to visualize structural abnormalities, such as the presence of hyper or hyporeflective lesions of specific layers, the presence of hemorrhage, fluid or exudates, pigment clumping, or choroidal masses. Quantitative indicators include measurements of the value of the reflectance and the gray scale difference between the lesion and contrasting normal tissue. These measurements can be used to track disease progression and response to treatment over time.

3.1.1. Spectral bands

Narrow ranges of wavelengths are used to capture imaged tissue.⁵ These bands may be selected based on specific features of interest in the tissue. Traditionally, the spectral bands were correlated with the colors in visible light including 450–500 nm (blue), 500–600 nm (green), 600–700 nm (yellow/orange), 700–800 nm (red), 800–900 nm (NIR).^{13,20} Blue light is used to capture abnormalities in retinal nerve fiber layer (RNFL) and vitreoretinal interface, while green is applied in the imaging of neurosensory retinal disorders. Yellow/orange is more potent in detecting lesion in the RPE–Bruch membrane choriocapillaris complex, and infrared wavelengths are more penetrative, and choroidal changes are more readily visible in this spectra.^{32,40,42} Fig. 2 shows an image set generated by a commercialized multispectral

camera (C2000, Shenzhen, China) with typical spectrum sections in a healthy eye.

Reflectance refers to the amount of light that is reflected by the imaged tissue. This parameter is used to measure the intensity of light that has interacted with the tissue. Reflectance depends on the balance between absorption by the pigment component of tissue versus reflectance of imaging spectra.³² Generally, when imaging with a peak wavelength of the pigmentary matter (hemoglobin/melanin), the pathology will appear as hyporeflective. If the imaging spectra encounters a reflecting interface (retinal fiber layer, vitreoretinal interface), it would be presented as hyperreflective. There are 3 forms of reflectance in MSI, namely *hyperreflectance*, *hyporeflectance*, and *isoreflectance*, which refer to the increased, decreased or unchanged spectral reflectance, respectively, in comparison to its surrounding normal tissue.⁴⁰ The reflectance which was calculated on the pixel-to-pixel basis. Briefly, the reflectance values at each pixel for a particular wavelength (λ_n) was determined using the recorded spectral images, while a calibrated white reference was used to account for differences in LED power and camera sensitivity across the spectral range. To measure the background noise, dark current images were taken without any eye present in the system, and immediately after each acquisition of a multispectral sequence of the ocular fundus. The ocular fundus, reference white and dark current images were all captured using the same exposure times for each spectral band. The reflectance values were then computed using standard techniques described in the literature.¹ In Table 1, we summarized the anatomical location, pathology, reflective pattern (hypo-, hyper-, and isoreflectance), and mechanism of reflectance of each chorioretinal disorders in clinical practice with its discrete discriminators based on literature research. Basic reflectance descriptions are also listed in Table 1. The terminology of reflectance is generally a qualitative indicator for the lesions due to lack of internal measurement tools in most of the MSI cameras so far. By calculating the value of the reflectance, it can be used as a quantitative indicator as stated below in reflectance index (RI).³⁰

3.1.2. Absorption spectrum

A graph that shows the amount of light absorbed by specific tissue at different wavelengths can provide information about the tissue's optical properties. The absorption spectrum was demonstrated in Fig. 1B.

Spectral signature refers to the unique pattern of reflectance values at different spectral bands for the pigmentary distribution in the fundus, which possessed characteristic pattern of light absorption and reflection by different structures. The spectral signature can be used to identify the pathology/lesion and distinguish it from other materials/normal tissue. For example, spectral difference (spectral signature) on the retinal appearance was found between Alzheimer disease patients and healthy controls based on MSI, in a recent study by Hadoux,¹⁸ allowing the early detection for Alzheimer disease by MSI.

Retinal transmission is a both a qualitative and quantitative indicator in MSI. Retinal transmission in MSI refers to the amount of light that is transmitted through the retina and

reaches choroid before finally being detected by the camera.³⁷ It is defined by the presence of choroidal vasculature in specific wavelengths. In general, the transmission of light through the retina depends on the absorption and scattering properties of the various retinal components, such as melanin, hemoglobin, and other pigments. Melanin, for example, absorbs light in the visible spectrum and can therefore affect the transmission of light through the retina.⁴² Hemoglobin, on the other hand, absorbs light in the NIR spectrum and can affect the transmission of light at these wavelengths.¹⁸ In healthy eyes, retinal transmission (or exposure of choroidal vessels) follows a specific order: it first takes place around the optic disc, particularly the inferior area of the optic disc. In red or NIR wavelengths, choroidal vessels are gradually exposed in the temporal regions of the macula until they finally emerge within the macular region. Retinal transmission provides reference exposure grades based on different axial length/age groups of healthy subjects.³⁷ The exposure grade of choroidal vessels could be used as an indicator of the transmission function of the retina using MSI. By measuring the transmission of light through the retina at different wavelengths, retinal transmission can provide qualitative and quantitative information about the distribution and concentration of various retinal components. For example, presence of retinal transmission at 550–600 nm was observed in atrophic chorioretinopathy secondary to osteoma and pathological myopia (Fig. 3).²⁸

3.1.3. Reflectance index

Quantitative measurement of reflectance is defined by the reflectance index, which is an important indicator in normal tissue and certain pathologies.³⁰ This quantitative parameter describes reflectance value of the lesion. Measurement is performed according to the previously reported protocol.³⁰ Briefly, five measurement sites were previously selected for measurements in the homogenous-reflectance lesion, namely the center, superior, inferior, nasal, and temporal sites. Gray scale readings were obtained by Image J (Rout: Analyze-Histogram) with black calibrated as 0 and white calibrated as 255. The RI was calculated as the average of the five measurements.

Border definition index (BDI) is another comparative measurement of the reflectance between lesions and normal tissue. BDI is defined as the average gray scale difference between the lesion and its surrounding retinal tissue.³⁰ Measurements were performed on ImageJ by selecting 12 sites in a clockwise pattern. The gray scale difference 100 on the linear 100 pixel across the lesion and the normal surrounding tissue was measured. So far BDI has shown advantages in differentiation of lesions compared with AF and FFA.³⁰

4. Clinical application

4.1. Retinal vascular disorders

4.1.1. Diabetic retinopathy

MSI was investigated for diagnostic value and staging accuracy for DR of various severity. In nonproliferative (NPDR)

Table 1 – Characteristics of reflectance spectra.

Imaging spectra	Anatomical location	Components	Pathology	Reflectance	Mechanism
550–650 nm	Vitreoretinal Interface	Hemoglobin	NVE/NVD	Hypo-reflective	Maximum absorption of hemoglobin around 580 nm
	Vitreoretinal Interface	Increased Interface-reflection	Epiretinal-membrane/Cotton wool spot	Hyperreflective	Reflection from the interface or lesion
	Vitreoretinal Interface	Decreased Interface-reflection	Macular hole/Retinal nerve fiber layer defect	Hypo-reflective	Lack of reflection from the vitreoretinal interface that results in decreased reflection/Lack of reflection due to nerve fiber defect
	Intraretinal	Hemoglobin	Hemorrhage/Microaneurysm/IRMA	Hypo-reflective	Maximum absorption of hemoglobin around 580 nm
	Intraretinal RPE	Lipid Water	Hard exudates Subretinal fluid	Hyperreflective Iso-reflectance	Decreased absorption of lipids Water presented minimal absorption over 500 nm with SRF showed iso-reflectance within the detached area
650–700 nm		Increased pigment absorption	RPE clumping/RPE hypertrophy	Hyporreflection	Increased pigment in PRE layer results in more potent absorption of the specter, leading to hyporeflectance
		Decrease pigment absorption	PRE dystrophy/RPE rip/ RPE defect	Hyperrefrection	Decreased PRE absorption due to decreased pigment in PRE
		Interface reflection	Pigment epithelium detachment (PED)	Hyperrefrection(ring-shape)	Hyperrefrection at the border of PED
	Sub-PRE	Hemoglobin	BVN/Polyps/CNV	Hyperrefrection	Hemoglobin do not absorb light over 600 nm, presenting vascular structure as brighter (hyperrefrection)
	Stroma	Melanin	Nevus/Choroidal tumor with melanin/Choroidal osteoma	Hypo reflective	Increased absorption by melanin in the stroma
700–850 nm		Increased Interface-reflection		Hyperreflective	Increase reflection from the interface of the osteoma
	Vascular	Hemoglobin	Haller's layer/Choroidal hemangioma	Hyperreflective	Hemoglobin showed no absorption while the stroma with melanin showed relatively more absorption

RPE = retinal pigment epithelium; NVE = neovascularization elsewhere; NVD = neovascularization of the disc; IRMA = intraretinal microvascular abnormalities; BVN = branching vascular networks; CNV = choroidal neovascularization.

Table 2 – Comparison of different types of multispectral (MSI) devices.

Technical device	Imaging spectrum (nm)	Bandwidth	Field of view (FOV)	Oxy-deoxy map	Advantages	Disadvantage
Multispectral retinal image analysis (MRIA)	507, 525, 553, 584, 596, 617	12 nm	45	NA	First investigation of MSI as a potential option as noninvasive tool in quantifying macular pigment, retinal hemoglobins and possibly other retinal and choroidal parameters in normal subjects. MRIA may be applicable to a number of clinical scenarios. Full spectrum mode and FastScan mode	The exposure time was too long and image movement artefacts were inevitably introduced; some of these artefacts cannot be compensated for by simple registering of the six images
RHA (Anmidis Health Systems, Ottawa, Canada) C2000	550, 580, 590, 620, 660, 690, 740, 760, 780, 810, and 850 550, 600, 620, 680, 780, 810, 850	10 nm 5 nm	42 63	Available NA	Large FOV, no-dilation required, high resolution	Dilation required, imaging time reported 10 minutes Lack of oxy-deoxy map

stages, microaneurysms are biomarkers of microvascular injury. The microaneurysms were defined by MSI as small blood vessel outpouchings and capillary dilations which have been observed clinically with 580 nm wavelength.¹⁷

Imaging principle: An imaging wavelength of 580 nm was chosen to identify the characteristics of retinal abnormalities in DR. This is mainly based on the reflectance and absorption characteristics of hemoglobin within the DR abnormalities, such as microaneurysms, intraretinal microvascular abnormality (IRMA), and neovascularization which were considered the major biomarkers for the progression for DR. Hemoglobin showed extensive absorption between 550 and 600 nm,⁴² presenting microaneurysm and hemorrhages within the retina as hypo-reflective lesions. On MSI microaneurysms presented as scattered, well-demarcated, round, hyporeflective spots, while retinal hemorrhages were displayed as larger areas of hyporeflective lesions with irregular shape. Hard exudates showed higher reflectance than cotton wool spots on MSI with the former presented with sharper borders. IRMA was defined as abnormally dilated microvascularization located near the main vascular trunks, whereas neovascularization elsewhere on MSI were depicted as a more disorganized network.²⁶ Photocoagulation spots were also frequently investigated in DR, and it showed characteristic features of hyporeflective border with minimal hyperreflective or iso-reflective center. Representative images of neovascularization of the disc, IRMA, microaneurysm, and hemorrhage spot are demonstrated in Fig. 4.

Related clinical research has been performed to investigate the sensitivity and specificity of MSI in diagnosis and grading of DR in multiple studies.^{17,26} A recent study by Liang and coworkers⁸ that included a total of 50 type-2 diabetic patients (99 eyes) showed the agreement between the FFA and MSI was 0.835, versus 0.614 for color fundus photo (CFP) and MSI, with FFA considered as gold. The sensitivity of no DR/mild NPDR in both MSI and CFP was 100%, and for severe NPDR/PDR it was 97.4% for MSI and 88.3% for CFP, respectively. The superiority of MSI in detecting MA and IRMA arises from the optical behavior of hemoglobin with the maximum absorbance of light by hemoglobin between 560 and 600 nm²⁷ that allows significant greater contrast in detection of the discriminators of DR of all stages. All these features were helpful in achieving good agreement with FFA in DR grading.

Advantages of MSI in DR lies in its high sensitivity and specificity of all stages of DR, allowing for DR screening, diagnosis, and staging in a noninvasive, multifaceted manner, with almost the same image acquisition timing as fundus photography.

4.2. Retinal vein occlusion (RVO)

MSI was applied in the evaluation of RVO by Xu and coworkers to reveal the characteristics of RVO by MSI.⁴³ The receiver operating curve analysis showed that MSI was a better discriminator of RVO than CFP (area under the curve = 0.911 vs. 0.768, $P = 0.0318$). MSI revealed highly defined vascular abnormalities in shortwave spectrum images and oxy-deoxy maps, which was compatible with CFP, FFA, and optical coherence tomography (OCT) findings. This demonstrated

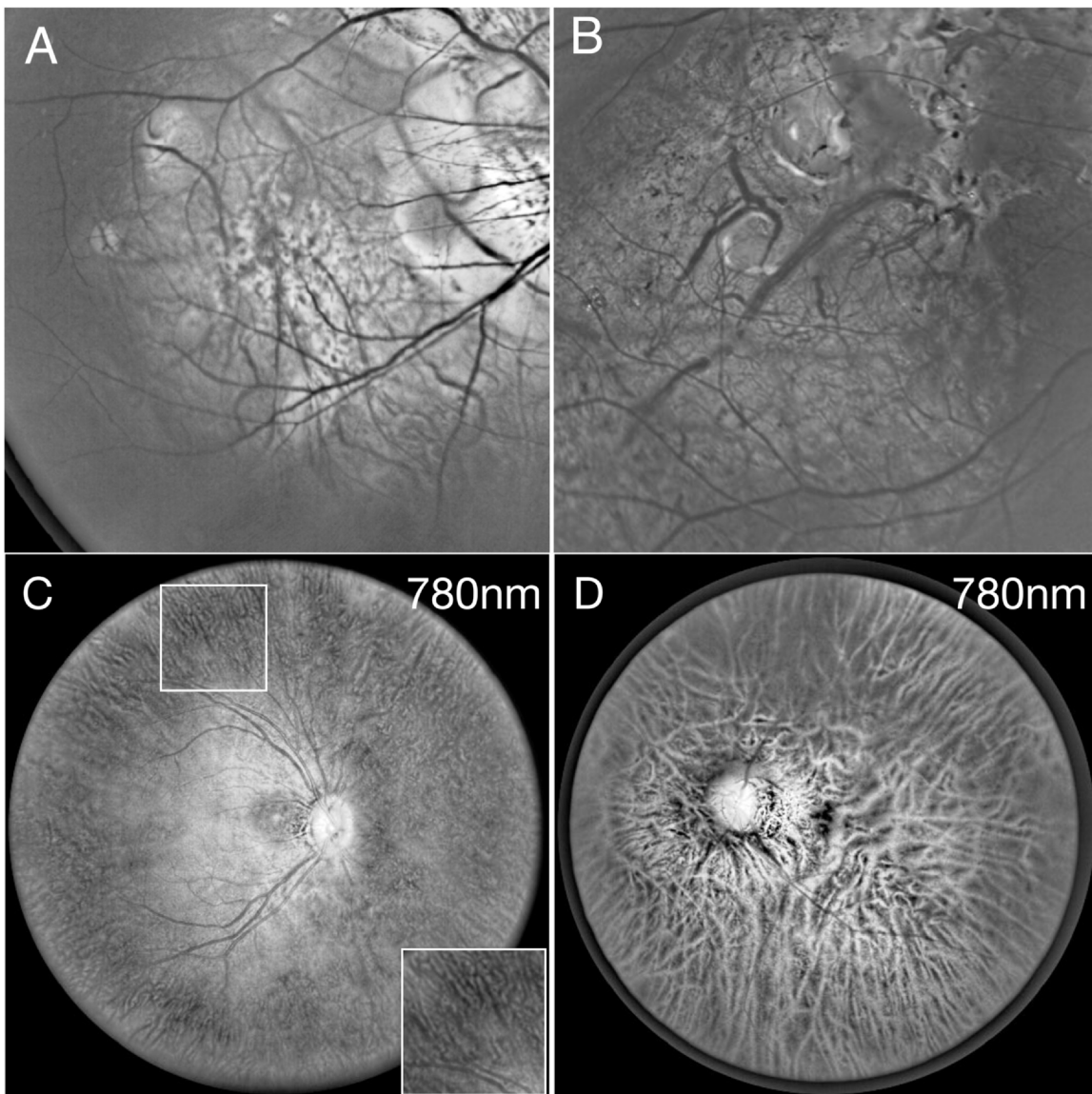


Figure 3 – Different forms of retinal transmission. A: Retinal transmission at 550 nm of choroidal vessels in high myopia showing increased backscattered reflectance from the choroid and shadowing effect of the large choroidal vessels. **B:** Retinal transmission in 550 nm in choroidal osteoma with chorioretinal atrophy showing choroidal vasculature exposure that exhibited as dark linear lines of different caliber in the macular area. **C:** Sattler's layer transmitted at 700 nm. **D:** Haller's layer transmitted extensively in the field of view of 63 degrees at 780 nm.

preliminarily the advantages of noninvasiveness, simplicity, and effectiveness of MSI in evaluating RVO pathology.⁴³ MSI allows for the circling of nonperfusion area when photo-coagulation is indicated in the treatment. Fig. 5 shows an example of MSI in RVO.

4.3. Age-related macular degeneration

Imaging of melanin disruption in RPE has been important because of its pathological significance in AMD. The imaging principle for RPE is based on the optical-chemical characteristics of melanin described by Prota.³⁵ Melanin absorption falls within the wavelength spectrum used by traditional fundus cameras, ranging from 450 to 700 nm, between which

melanin is obscured by other dominant components of the eye including the lens, hemoglobin, zeaxanthin, lutein, and rhodopsin. Using middle to long wavelengths of MSI (such as 680 nm), has revealed that the back reflection from the sclera becomes more intense and the visibility of melanin disruption is more apparent. Melanin disruption was defined as increased pigmentation, particularly melanin clumping, stacking, or migration but not specifically atrophy. The results of a study by Dugel and coworkers¹⁰ indicated that 51 of 61 eyes (84%) of 43 patients with AMD were found to have melanin disruption in MSI images in at least the central and/or one of 4 inner Early Treatment Diabetic Retinopathy Study circle areas. The sensitivity of fundus photography for melanin pigment as compared to MSI was only 62.5%, with 3 false

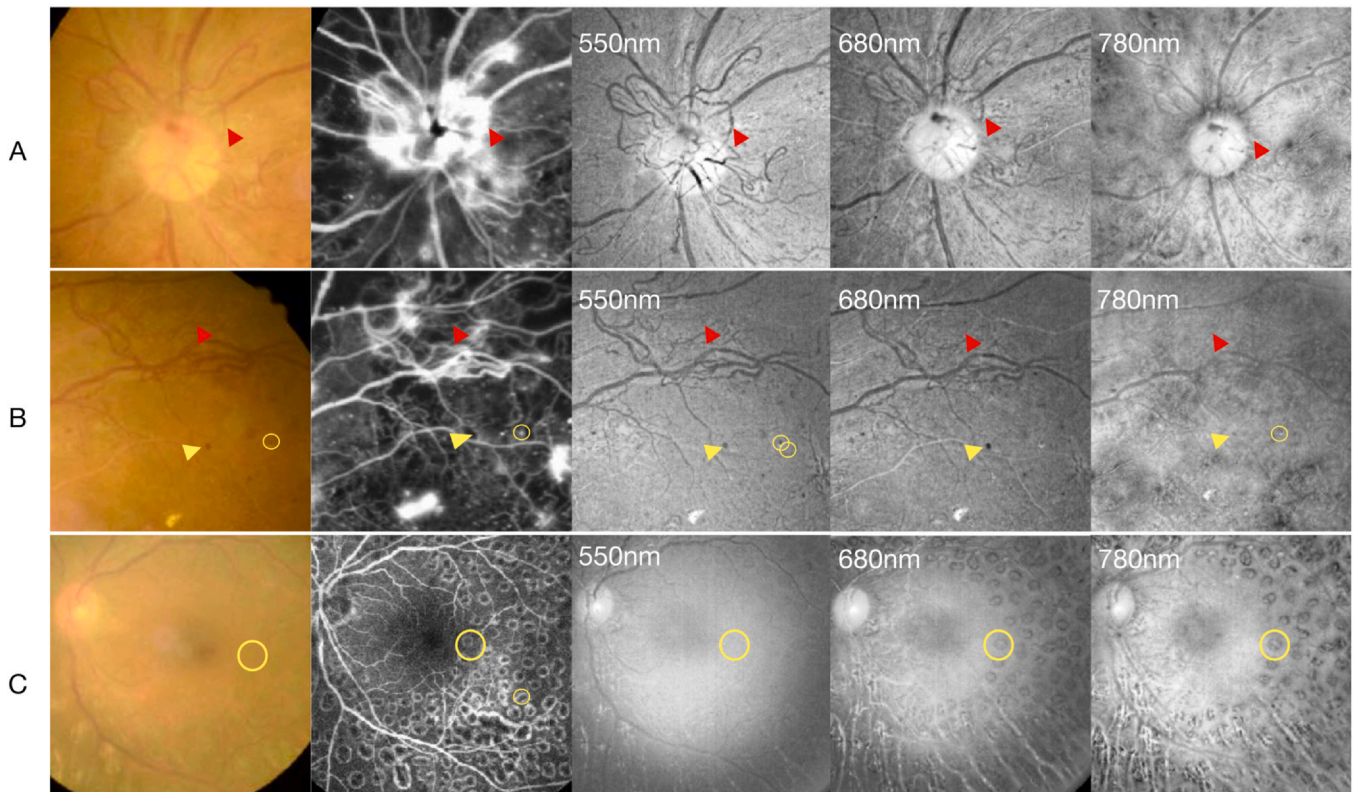


Figure 4 – Multispectral imaging of high-risk proliferative diabetic retinopathy with neovascularization. A: Neovascularization of the disc (red arrow). B: Neovascularization elsewhere (red arrow) and diabetic microaneurysms (yellow arrow). C: Panretinal photocoagulation for proliferative diabetic retinopathy (yellow circle).

negatives; however, there was a lack of confirmation by other imaging modalities, including OCT in validation of the melanin migration or clumping on sectional views. In addition to dry AMD, RPE disruption around wet AMD lesions was also considered to be an important implication for wet AMD. MSI can aid in detection of early RPE atrophy after anti-vascular endothelial growth factor (VEGF) treatment in patients diagnosed with wet AMD.²¹ This was observed by Zhang and coworkers who stated that MSI could detect the *de novo* sign of circular hyperreflectivity around the choroidal neovascularization (CNV) lesion post-treatment, giving it a pattern of bull's-eye pattern of RPE atrophy in NIR spectrum before and after anti-VEGF treatment. This was particularly prominent at middle to long wavelengths. Interestingly, this circular atrophy signal was relatively weak on OCT scan where subtle alternations in RPE morphology may not be detected due to normal RPE and adjacent atrophic RPE or melanin disruption.²¹

Fig. 6 shows an example of AMD clinical spectra using MSI. In CNV there is a decrease in optical absorption by hemoglobin in the CNV after 600 nm, presenting CNV lesions as hyper-reflective in the macular area (Fig. 6 Row B). Drusen present as solitary or confluent hyper-reflective lesions with blurry border in short wavelengths. The border appears more defined in middle to long wavelengths due to more penetration into outer retina in long wavelengths.

MSI seems to show a high detection rate for PRE disruption, yet its real-world experience and diagnostic value in CNV is still under further exploration.

4.4. Polypoidal choroidal vasculopathy

Polypoid lesions and branching vascular networks (BVN) are defining features for PCV. Polypoidal lesion in MSI was further described as a polyp-like cluster of hyper-reflectance in the short-wavelength images with subsequent highly defined ring-like hyper-reflectance in the longer-wavelength images.²⁹ BVN is also characterized in MSI as patchy hyper-reflectance composed of intertwining lines with well-defined borders in 680 nm and above. The hypothesized mechanism is that hemoglobin is more reflective for wavelengths over 600 nm, leading to more reflectance from BVN that originated from the choriocapillaris.²⁹

Relevant clinical studies have been performed to investigate sensitivity and specificity of MSI PCV diagnosis.^{22,29,45} In a small case series of 14 patients, the characteristics of polypoidal lesions and BVN were initially investigated and compared with ICGA.⁴² Detection rate for polypoidal lesion was measured as primary outcome in the investigated patients, showing 15 out of 16 eyes presented with the predefined lesion. Diagnostic value for PCV on MSI was performed in Ma's study that included 67 eyes.

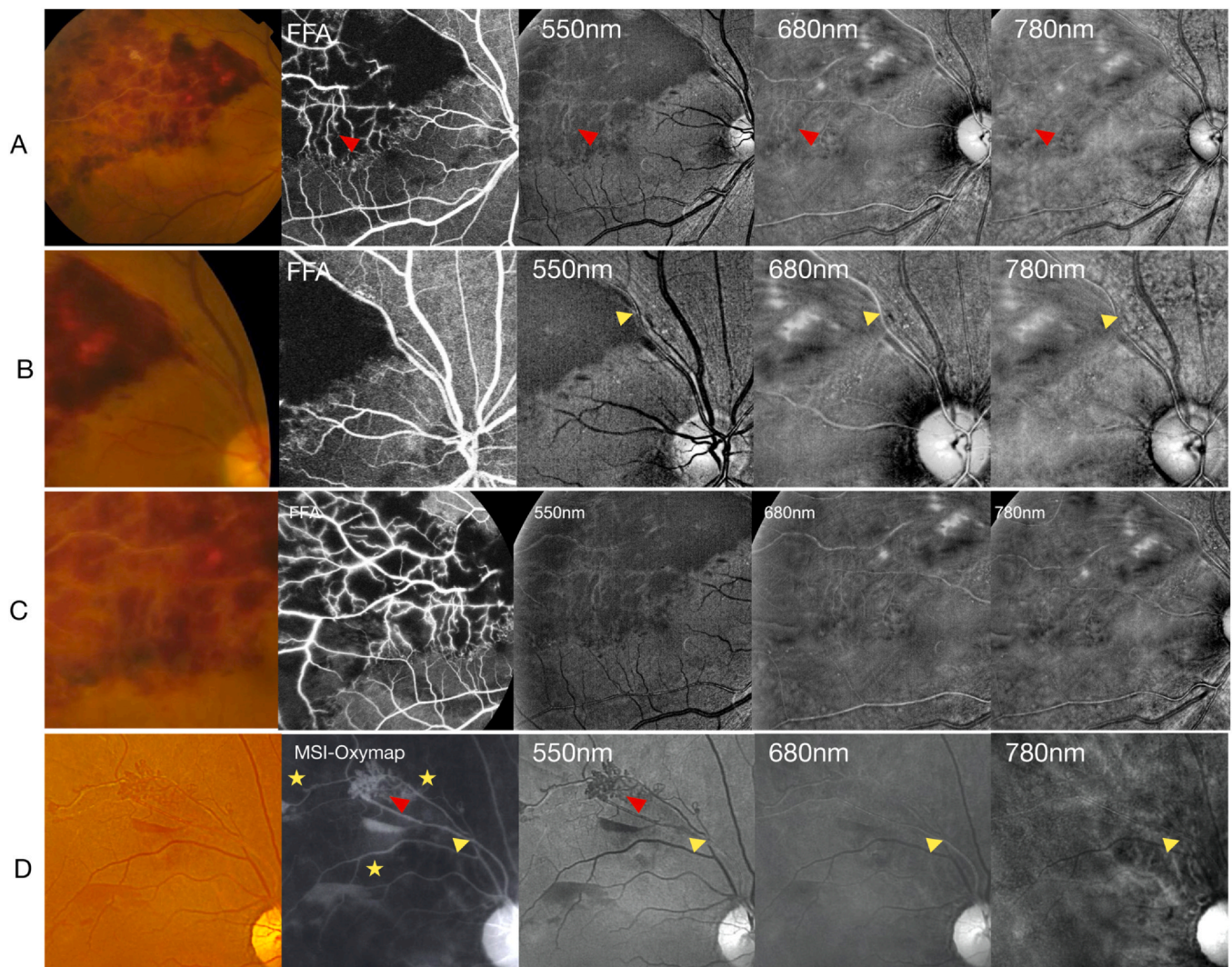


Figure 5 – Multispectral imaging showing hemiretinal vein occlusion. A–C: Acute hemorrhage and recurrent bleeds (red arrowhead) showing the areas of retinal edema and nonperfusion (yellow arrowheads). Hyper-reflectance is seen at the 550 nm due to presence of hemorrhage. Retinal edema and cotton-wool spots are seen at the 680 and 780 nm. D: Resolved hemorrhage showing ischemic neovascularization at 550 nm (red arrowheads). Deeper structure and occluded retinal vessels are seen at longer wavelengths.

Compared with ICGA, sensitivity and specificity for diagnosing PCV by MSI alone was 84.21% and 93.10%, respectively.²⁹

Advantages/disadvantages and future applications: MSI allows noninvasive visualization of polypoidal lesions and BVN and potentially serves as a new diagnostic imaging option for PCV. Other noninvasive imaging tools such as multicolor has also been validated in PCV.⁴¹ In a cross-sectional study, polyps were described as dark gray oval lesions with distinct margins most clearly seen on the infrared reflectance images of multicolor, with a detection rate of 98%. On the multicolor composite images, polyps appeared as dark green oval lesions. The BVN appeared as mottled gray regions on infrared reflectance imaging and were seen less frequently compared with polyps (30/50 eyes, 60%), though the diagnostic value of multicolor on PCV has not been investigated.

The consistency that longer wavelengths were able to reveal the presence of polypoidal lesions both in MSI and multicolor indicated the impactful spectral reflectance from deeper tissue in the retina and the choroid. However, diagnostic performance on an even larger cohort is still to be investigated.

4.5. Punctate inner choroidopathy (PIC)

MSI have been recently applied in the observation of PIC by Zhang.⁴⁶ In general, PIC lesions were observed as small, circular structures with high reflectivity on MSI; however, PIC lesions present differently at various stages. In cases of atrophic and punched-out lesions, they tended to be particularly reflective due to RPE atrophy across all wavelengths, giving them a pale appearance, and may be accompanied by

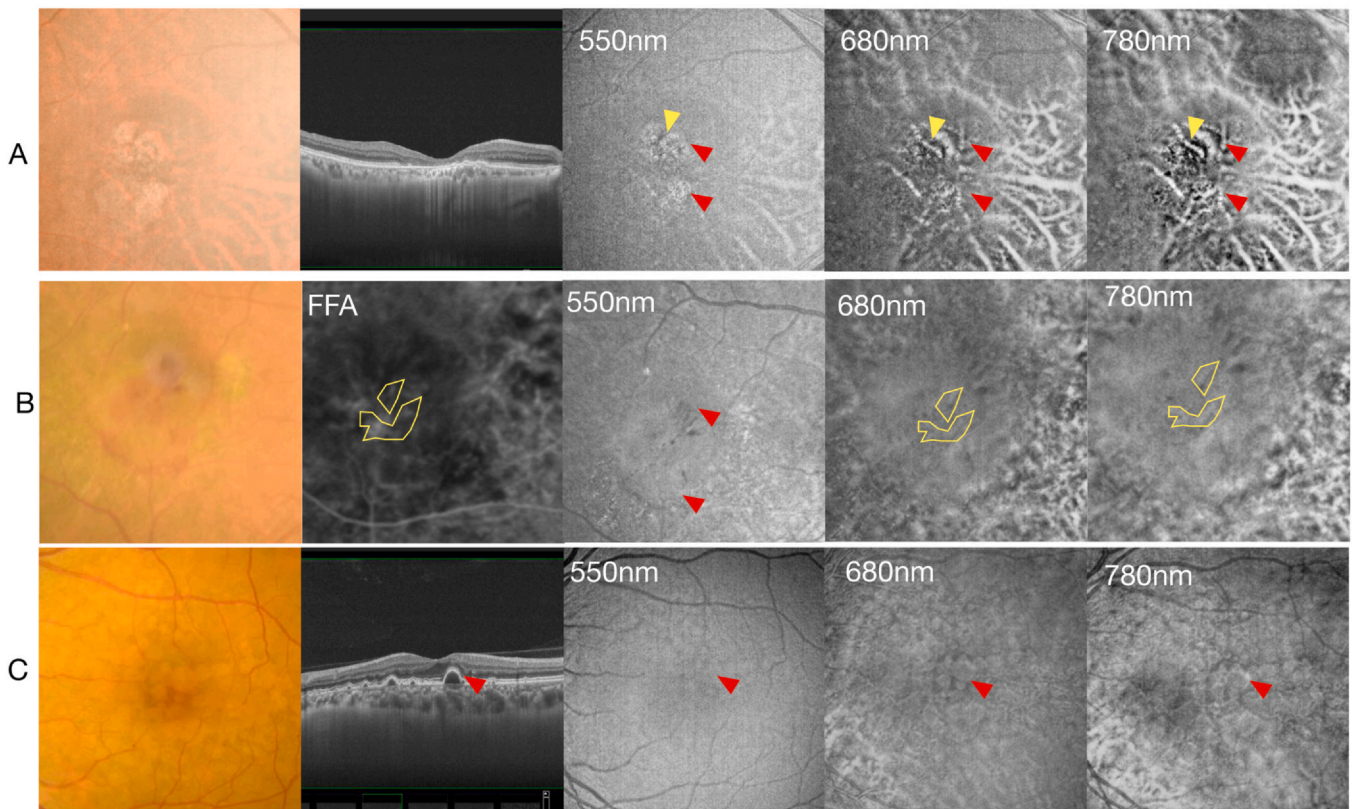


Figure 6 – Multispectral imaging of different biological changes of age-related macular degeneration (AMD). A: Geographic atrophy with retinal pigment epithelium disruption. B: Exudative AMD with hyperreflective lesions in the macula in 680 and 780 nm (yellow encircled area). C: Drusen presented as minimal hyperreflective lesion with blurry borders at 550 nm, with sharper borders at 680 and longer wavelengths (red arrowheads).

nonreflective melanin stacking at the edges or appearing mottled and dark in appearance; however, newly formed lesions were relatively less reflective, and could only be visualized on MSI images captured at longer wavelengths ranging from 620 to 850 nm. MSI showed excellent detection of CNV lesions secondary to PIC. Advantages of MSI still lies in its CNV diagnosis; however, MSI failed in detecting all PIC lesions, with a detection rate of 58.3%. This is a reminder that in active inflammatory lesions such as active PIC, MSI may not provide enough diagnostic information when RPE is still not intact.⁴⁶

4.6. RPE disorders and uveitis

4.6.1. RPE disorders - central serous chorioretinopathy (CSCR)

The diagnosis for CSCR, though easily made clinically and by OCT, is still a target for the localization of RPE defect. This used to be identified by FFA, showing smoke-stack leakage. The identification of leakage on MSI presents as hyper-reflective signal due to weak absorption, whereas hyperpigmented RPE presents as hypo-reflective signal.⁴⁸ This is from decreased PRE tight junction that may be related to weaker melanin distribution. Zhu and coworkers have found that on MSI, RPE leakage was observed in 25 of 26 eyes (96.2%) with

CSCR, which was comparable to the performance of FFA.⁴⁸ Dome-shaped areas of signal, which corresponded to fluid accumulation associated with neurosensory retinal detachment or RPE detachment were observed in 23 of 26 eyes (88.5%). This could potentially aid in noninvasive and dyeless localization of leakage and further laser treatment guidance; however, there is still lack of sufficient evidence in a large study population in pinpointing all active sites by MSI, which is vital when laser treatment is indicated.

4.6.2. Vogt-Koyanagi-Harada disease (VKH)

Key indicators for the clinical diagnosis of VKH have relied on the use of FFA to detect bilateral ocular involvement with diffuse choroiditis, exudative retinal detachment, RPE changes, or disc edema. OCT has provided additional diagnostic information.⁴⁵ Imaging principle of MSI on the mentioned indicators keeps consistent with the previously mentioned pathologies including neurosensory detachment and RPE disruption.²⁹ Clinical diagnosis of VKH has relied on the use of FFA. The application of MSI in VKH has added to our knowledge of the RPE changes in all stages of VKH, with more sensitivity in detecting RPE changes in VKH than other imaging modalities.^{14,19} Clinical relevance/advantages of MSI in VKH lie in the discovery of pronounced presentation for pigmentation/depigmentation not seen with other imaging modalities including OCT and fundus AF.; however, there is

still lack of data as to the presentation when VKH subsidized, including decreased neurosensory detachment and choroidal thickness on OCT.

4.6.3. Hydroxychloroquine (HCQ) toxicity

HCQ toxicity can result in bull's-eye pattern of atrophy, although deeper penetration with scans in longer wavelengths is necessary to illustrate this pathologic change. In a case report of HCQ toxicity by Bhavsar and coworkers,⁴ MSI showed features of bull's eye pattern of RPE loss that was evident at higher wavelengths with deeper penetration such as 740 nm but not in shorter wavelengths such as 580 nm. This pilot case report imaged by MSI showed its relevance in enabling targeted evaluation of different retinal layers through spectral and depth-enhanced differential visibility such as in RPE defects induced by HCQ. AF has been considered a sensitive imaging tool for the early detection of HCQ toxicity,⁴ though comparison between MSI and AF has not yet been studied.

4.6.4. Stargardt disease

In a pilot case report of Stargardt disease imaged by MSI, it was revealed that perifoveal flecks are seen on the longest (940 nm) and shortest (580 nm) wavelengths, which showed better enhancement compared with CFP. In cases of early or mild changes of Stargardt disease, visualization of subtle flecks may be helpful in making an early diagnosis. The clinical relevance of MSI, therefore, lies in its ability to provide early biomarker for precise diagnosis.³⁴

4.6.5. Choroideremia

MSI significantly improves visualization of the RPE in choroideremia.⁹ MSI-620 nm through MSI-660 nm images show increased visibility of pigment mottling and RPE disruption with an increase in wavelength and depth of penetration. With the use of MSI and the specific selection of spectral bands, the subretinal structures and metabolic features of the retina and choroid can be visualized in finer details. This makes MSI a valuable tool for diagnosing and monitoring the effect of retinal and choroidal preservation of patients with choroideremia in combination with *en face* high-resolution OCT.

4.6.6. Choroidal tumors

(1) Neurofibromatosis type 1 (NF-1)

The abnormalities of the choroid have been hypothesized to correspond with neurofibromas, hamartomas, areas of choroidal ischemia, café au lait spots, nevi, or simple choroidal nodules. Imaging of the nodules was almost impossible due to the blockage from PRE layer and choroidal pigment. In one case, scans in the red and infrared spectrum detected numerous deep choroidal lesions, which showed better sensitivity than other imaging modalities.⁷

(2) Choroidal hemangioma (CH)

Hemangioma on MSI is characterized by circumscribed choroidal lesions with loss of vasculature details from 590 to 850 nm in contrast to normal choroidal vascular details of the remainder of the fundus.¹⁶ Transscleral choroidal imaging TCI with 940 nm shows the disorganized vascular

network within the CH. In comparison, mid-phase indocyanine green angiogram and MSI choroidal oxy-deoxyhemoglobin contrasted map (OH2) images both demonstrate hyper-reflectance of the lesion, indicating oxygenated blood within the CH.¹⁶ Clinical relevance is in its value in revealing both the morphological and oxygenated information within the CH. Whether the spectrum characteristics can be applied as differential diagnostic features needs further investigation.

(3) Choroidal osteoma

MSI detects different reflectance in calcified and decalcified osteomas. In a case series by Ma and coworkers,²⁸ calcified choroidal osteoma exhibited homogeneous reflectance in 550 nm with the beehive appearance in 600–680 nm and homogenous hyper reflectance 780–850 nm. This possibly indicated the compact bone in the outer layers and bone trabecula in the middle layer, which is highly corresponding to the OCT features. This was named as a “sandwich sign.” MSI was able to differentiate the inactive portion of the osteoma from the decalcified area. The inactive portion was characterized by geographic hyper-reflective islands with higher reflectivity border (“floating island sign”), while decalcified portion was featured by increased definition and reflectivity from osteoma. Clinical relevance of MSI is in its ability in contributing to *en face* visualization of choroidal osteomas at different stages, providing new insights into the spectral behavior of choroidal osteoma.²⁸

Based on the published observations on different spectrum behavior of choroidal tumors, quantitative analysis of the reflectance value (reflectance index, RI) and the BDI were recently analyzed (unpublished data). While RI referred as average reflectance gray scale, BDI was defined as the average gray scale difference between the tumor and its surrounding retina tissue. Fig. 7 shows three types of choroidal tumors and choroidal nevus that were studied. Graph 1 presents the results showing different reflectance patterns in osteoma (A), CH (B), nevus (C), and melanoma (D). The border also showed differences between imaging modalities (Graph H). The application of quantified MSI in differential diagnosis still requires further validation, but the pilot study has already provided a clue regarding its value in choroidal tumor diagnosis. Table 3 compares sensitivity, specificity, and area under the curve in clinical entities reported above.

5. Future trends in MSI

MSI has the potential to revolutionize the diagnosis and treatment of various eye diseases. Some future trends in MSI in ophthalmology include:

- (1) *Projection-removal technique*: Projection removal is the extraction of specific slice(s) of the fundus image that allows more detailed analysis and tissue-specific optical reflectance.
- (2) *Integration with other technologies*: MSI is likely to be integrated with other technologies, such as spectral-domain OCT and adaptive optics, in order to provide a more comprehensive view of the retina. This could help to identify disease processes at an earlier stage.

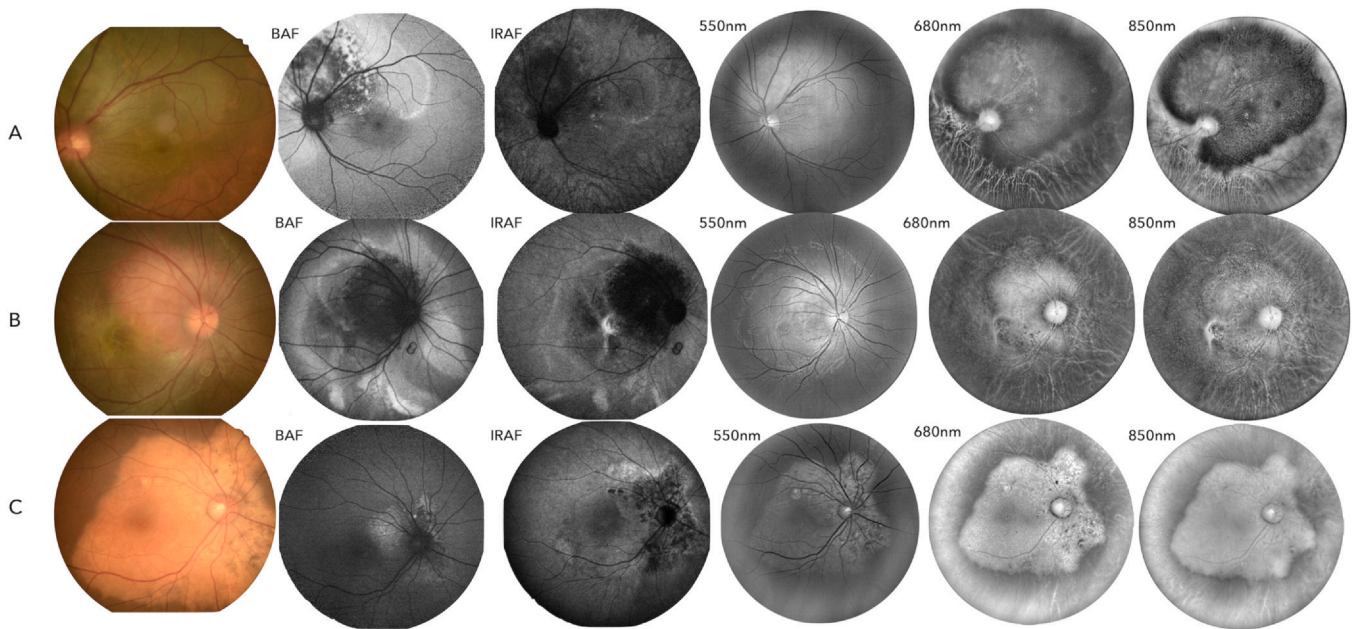
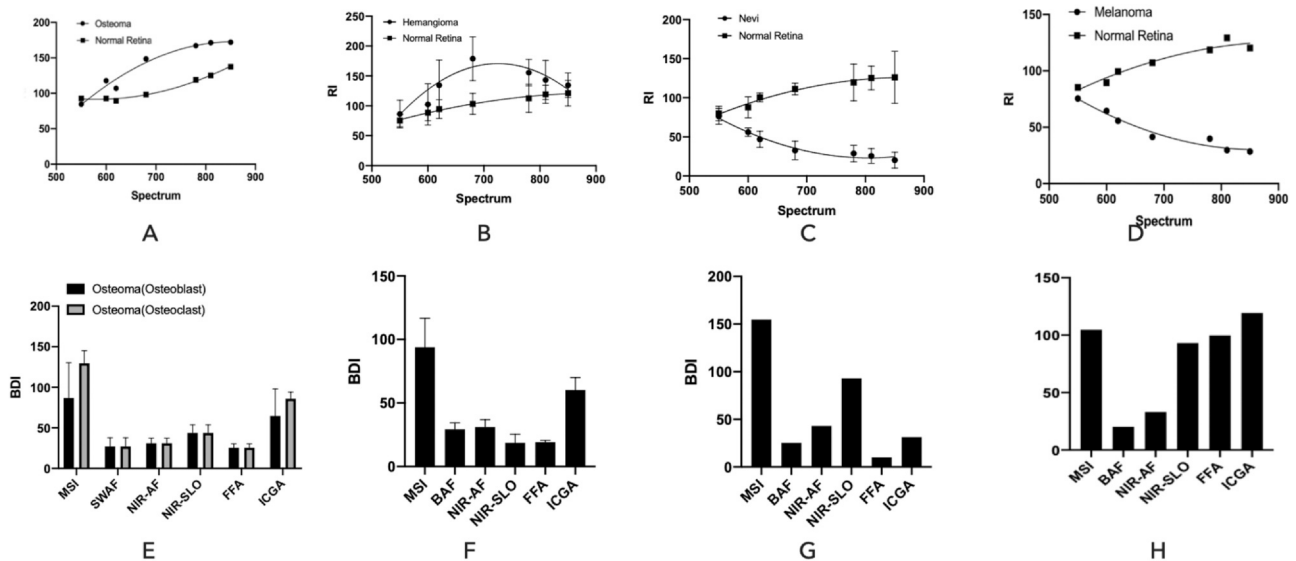


Figure 7 – Multimodal imaging of different types of choroidal tumors including choroidal melanoma (A), choroidal hemangioma (B), and choroidal osteoma (C). Blue autofluorescence and infrared autofluorescence provide information of secondary retinal pigment epithelium disruption from tumor, but both imaging modalities failed to show the presence or the border of the tumors. Multispectral imaging, selected from short (550 nm), middle (680 nm), and long-wavelengths (850 nm), presents the choroidal tumors in different reflectance patterns. In choroidal melanoma, the tumor shows increased hyporeflectance from 550 to 850 nm. The reflectance of hemangioma reaches its maximum in 680 nm but decreases in 850 nm, with interlaced pattern within the tumor. While in osteoma, the reflectance increased in intensity with honeycomb pattern within the tumor.



Graph 1 – Quantitative analysis for different types of choroidal tumors. Osteoma showed darker reflectance compared with its surrounding tissue (A). Hemangioma showed brighter reflectance than its surrounding tissue, but with its maximum intensity around 680–700 nm (B). Both choroidal nevus (C) and choroidal melanoma (D) showed decreased intensity in reflectance, with choroidal nevus showed significantly lowered reflectance than melanoma in 680–850 nm ($P_{680} = 0.013$, $P_{780} = 0.003$, $P_{850} = 0.001$). Border definition index were also compared in different imaging modalities. Multispectral imaging was able to present the border of osteoma, either osteoclast or osteoblast, which was significantly higher than autofluorescence/fundus fluorescein angiography (E). The same phenomenon was found in other choroidal tumors including hemangioma (F), nevus (G), and melanoma (H). Similar border contrast can be found in indocyanine green angiography.

Table 3 – Summary of sensitivity, specificity, and area under the curve (AUC) in reported clinical entities.

Clinical entity	N (eyes)	Discriminators	Sensitivity	Specificity	AUC
DR(NPDR)	44	-	100%	96.3%	-
DR(PDR)	45	-	97.4%	100%	-
CSCR	-	-	84.6%	100%	-
PCV	67	Polyps	84.21% (72.06–96.36)	93.10% (83.29–102.91)	0.89 (0.80–0.97)
		BVN	95.83 (87.21–104.45)	88.37 (78.39–98.35)	0.92 (0.85–0.99)
		SRF	76.92 (65.08–88.77)	86.67 (67.18–106.15)	0.82 (0.70–0.94)
		PED	91.11 (82.46–99.76)	90.91 (77.86–103.96)	0.89 (0.80–0.99)
RVO	56	NPA	42.3% (18.8–70.4)	100.0% (89.5–100.0)	0.714
		Retinal Hemorrhage	80.0% (51.4–94.7)	97.6% (85.6–99.9)	0.888
		Cotton Wool Spot	90.0% (54.1–99.5)	97.8% (87.0–99.9)	0.939
		Hard Exudate	90.9% (57.1–99.5)	97.8% (86.8–99.9)	0.943
		Intraretinal Edema	21.1% (7.0–46.1)	100.0% (88.3–100.0)	0.605

DR = Diabetic Retinopathy; NPDR = non-proliferative diabetic retinopathy; PDR = proliferative diabetic retinopathy; CSCR = central serous chorioretinopathy; PCV = polypoidal choroidal vasculopathy; RVO = retinal vein occlusion.

(3) *Novel imaging signs*: MSI may still uncover novel observations not yet published. Though choroidal transmission was observed by Li and coworkers²⁵ a more detailed reflectance from choroidal vessels has not yet been published. The Sattler layer was defined as the layer of medium-sized choroidal vessels that were visualized as medium-sized, hypointense spaces surrounded by a hyper-intensive stroma observed on OCT.⁴⁷ This phenomenon could be elegantly observed on MSI with the Sattler layer seen as hypo-reflectance surrounded by hyperreflectance stroma. The Haller layer was defined as the outer choroidal layer containing the large choroidal vessels with reflectance shown on MSI. A major criterion to differentiate between Sattler layer and Haller layer was the density of the intervascular tissue, including the choroidal melanocytes, which contributed to the scattering of both the OCT and MSI signal within the choroidal stroma⁴⁷; however, the asymmetrical presence of Sattler's layer and Haller's layer on MSI still warrant investigation.

6. Limitations and perspectives

Limitations of the MSI technique include lack of precise correspondence observed in layer-by-layers retina and choroid. MSI is not a replacement for FFA. Detection of all lesions of retinal vasculopathy by MSI is relatively lower if compared with FA such as microaneurysm in DR and regression of minimal neovascularization elsewhere after PRP. We have added this limitation to the corresponding section. Because of hardware limitation, MSI devices usually cannot achieve high-spatial-resolution.⁴² Also due to limited understanding of the spatial distribution of chromophores in neurosensory retina (if any), detailed slicing of MSI is hardly done. Theoretically, hyperspectral imaging is a suitable method for a variety of applications by producing the spatial distribution of spectral signatures of all chromophores in the tissue.¹² The terms “hypercubes” or “data cubes” refer to the amount of data obtained by hyperspectral imagery form 3D structures consisting of the spatial variation corresponding to each of the wavebands of the hyperspectral sensor²³; however, hyperspectral imaging has

also failed to slice all spatial layers of retina, but rather it also focused on the known chromophores in the fundus.^{19,48}

7. Conclusion

We have performed a review of the literature to evaluate the application of MSI in ophthalmology. The published literature demonstrates that the different spectra of MSI have diagnostic value in chorioretinal disorders with less need for invasive tests. The oxy/deoxy map is useful in evaluating metabolic status of the lesions. Several studies on application of MSI are ongoing.

8. Literature search

A structured search of the Medline, PubMed, Web of Science, Library of Congress databases was conducted using the key words multispectral imaging, retinal imaging, choroidal imaging, fundus autofluorescence, fluorescein angiography, monochromatic camera. The languages included English, Chinese, French, and German. Additional references were culled from the bibliographies of these references. References were collected up to July 31, 2022. These references were evaluated for their pertinence to the topic, with special consideration given to papers that detailed principles of MSI. Articles that were repetitious or decidedly nonclinical were omitted from consideration as well as articles on multimodal and multicolor imaging.

9. Disclosure

None.

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None.

Declaration of Competing Interest

In the interest of transparency, we ask you to disclose all relationships/activities/interests listed below that are related to the content of your manuscript. “Related” means any relation with for-profit or not-for-profit third parties whose interests may be affected by the content of the manuscript. Disclosure represents a commitment to transparency and does not necessarily indicate a bias. If you are in doubt about whether to list a relationship/activity/interest, it is preferable that you do so.

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